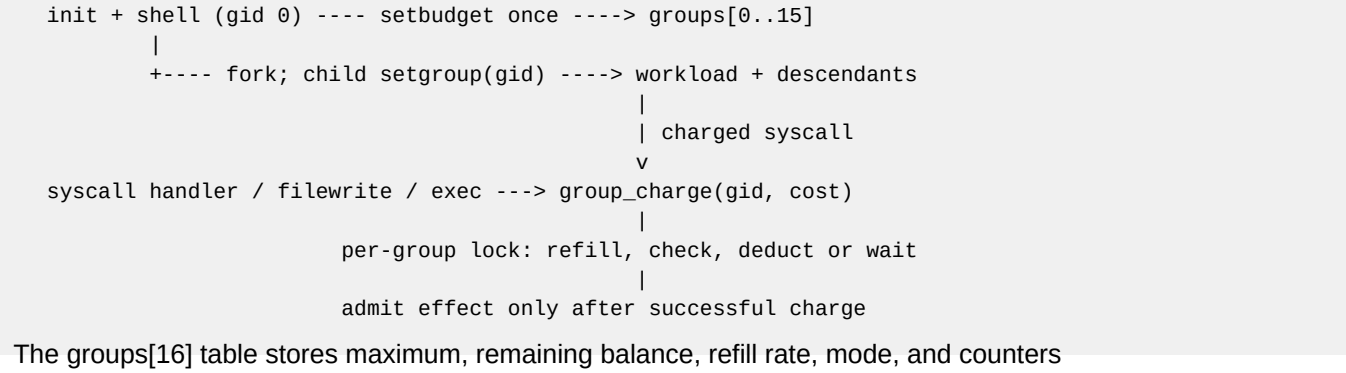
UndoBudget: Irreversibility Budget for xv6 Process Groups

CSL3030 project report Team: [Add names and roll numbers] Submission date: [Add date]

1. Goal and architecture

UndoBudget limits selected irreversible actions in the course-provided xv6-riscv tree. Processes in the same group spend from one shared balance. Fork copies a group ID, and exec preserves it; neither action creates a fresh budget. Group 0 is the trusted administrator. init configures group 0 and default workload group 1 once at boot. The shell stays in group 0 and moves command children into a configured workload group before redirection or exec.



The groups[16] table stores maximum, remaining balance, refill rate, mode, and counters under one spinlock per group. A process stores only gid. setbudget is group-0-only and succeeds once per group; workload membership is one-way. budgetstat returns a locked snapshot to user space.

2. Cost policy and enforcement

The group chose a cost policy because the PDF's numbers were examples. One positive-cost syscall is charged before its effect, with no refund after a later failure. A rejected operation has no intended irreversible effect.

Operation	Cost
unlink	12
Console write, n > 0	2 + ceil(n/128)
Same-group file write, n > 0	2 + ceil(n/1024)
Cross-group file write, n > 0	6 + 2*ceil(n/1024)
open(O_CREATE), open(O_TRUNC)	4, 10; both flags cost 14
mkdir / mknod, link, user exec	6, 3, 4
Other calls	0

The creator group is stored in both disk and memory inodes and survives reboot. It is initialized when an inode is allocated and cleared on reuse. This lets filewrite distinguish same-group and cross-group writes. The disk format and mkfs were updated together. New images contain 4,000 blocks so the additional programs leave space for xv6's writebig test.

3. Concurrency, challenges, and approach

FAIL mode refuses an unaffordable charge. BLOCK mode waits for enough points from full 100-tick refill periods, then rechecks and deducts atomically. The charge path takes tickslock before the group lock. A waiter registers on the tick channel before releasing the tick lock, preventing a lost wakeup; it sleeps with no spinlock held. Filesystem operations that might BLOCK release inode and log locks before waiting and repeat validation on retry.

Three practical problems shaped the implementation. First, this xv6 uses sleep_prepare()/sleep() and size-aware copyin/copyout, so upstream code could not be copied directly. Second, a console charge makes very small workloads unable to print their own failure. A trusted shell relay runs the workload in its selected group, forwards its pipe output, and charges group 0 for console output. Third, the larger user-program image initially caused writebig to run out of blocks; increasing new-image capacity fixed it.

A deterministic Python agent validates six allowlisted requests and sends them through a framed xv6 helper. File content is sent as bounded hex data, never as shell text. The bridge requires a matching completion marker and does not report success from an echo, timeout, or broken connection.

4. Verification and limits

On fresh images, the regression suite reported 32 pass, 0 fail, 0 untested across one and three CPUs. It covers authorization, fork/exec sharing, exact costs, refusal with surviving files, refill and BLOCK races, inode persistence, and agent responses. Ten repeatable experiments produced raw serial logs, CSV observations, and four plots. Full xv6 usertests and crash recovery also passed in a disposable copy. The 26-command demo sequence passed.

xv6 has no user identities or file permissions, so the group-0 shell and its binary are trusted. A failure after an admitted truncate or partial write can leave an incomplete file; the charge is retained by policy. Group slots are configured once per boot. These are project limits, not guarantees of general-purpose OS isolation.